AI Engineer

Introduction

- What is an AI Engineer?
  - Impact on Product Development
  - Roles and Responsiblities
- AI Engineer vs ML Engineer
- Common Terminology
  - AI vs AGI
  - LLMs
  - Inference
  - Training
  - Embeddings
  - Vector Databases
  - AI Agents
  - RAG
  - Prompt Engineering

Using Pre-trained Models

Pre-trained Models

- Benefits of Pre-trained Models
- Limitations and Considerations
- OpenAI Models
  - Open AI Models
  - Capabilities / Context Length
  - Cut-off Dates / Knowledge
- Popular AI Models
  - Anthropic's Claude
  - Google's Gemini
  - Azure AI
  - AWS Sagemaker
  - Hugging Face Models
  - Mistral AI
  - Cohere
  - Replicate

Open AI Platform

OpenAI API

- Chat Completions API
- Writing Prompts
- Managing Tokens
  - Maximum Tokens
  - Token Counting
  - Pricing Considerations
- Open AI Playground
- Fine-tuning
- Prompt Engineering Roadmap

AI Safety and Ethics

- Prompt Injection Attacks
- Security and Privacy Concerns
- Bias and Fairness
- Understanding AI Safety Issues
- OpenAI Moderation API
- Adding end-user IDs in prompts
- Conducting adversarial testing
- Robust prompt engineering
- Know your Customers / Usecases
- Constraining outputs and inputs
- Safety Best Practices

OpenSource AI

- Open vs Closed Source Models
- Popular Open Source Models
- Hugging Face
  - Finding Open Source Models
    - Hugging Face Tasks
    - Hugging Face Hub
  - Using Open Source Models
    - Inference SDK
    - Transformers.js
- Ollama
  - Ollama Models
  - Ollama SDK

Embeddings & Vector Databases

What are Embeddings

- Semantic Search
- Data Classification
- Recommendation Systems
- Anomaly Detection
- Use Cases for Embeddings

Open AI Embeddings API

- Open AI Embedding Models
- Pricing Considerations

Open-Source Embeddings

- Sentence Transformers
- Models on Hugging Face

Vector Databases

- Purpose and Functionality
- Popular Vector DBs (pick one)
  - Chroma
  - Pinecone
  - Weaviate
  - FAISS
  - LanceDB
  - Qdrant
  - Supabase
  - MongoDB Atlas
- Implementing Vector Search
  - Indexing Embeddings
  - Performing Similarity Search

RAG & Implementation

- RAG Usecases
- RAG vs Fine-tuning
- Implementing RAG
  - Chunking
  - Embedding
  - Vector Database
  - Retrieval Process
  - Generation
- Ways of Implementing RAG
  - Using SDKs Directly
  - Langchain
  - Llama Index

RAG Alternative

- Open AI Assistant API

AI Agents

- Agents Usecases
- Prompt Engineering
- ReAct Prompting
- Building AI Agents
  - Manual Implementation
  - OpenAI Functions / Tools
  - OpenAI Assistant API

Multimodal AI

- Multimodal AI Usecases
  - Image Understanding
  - Image Generation
  - Video Understanding
  - Audio Processing
  - Text-to-Speech
  - Speech-to-Text
  - Multimodal AI Tasks
- OpenAI Vision API
- DALL-E API
- Whisper API
- Hugging Face Models
- LangChain for Multimodal Apps
- LlamaIndex for Multimodal Apps
- Implementing Multimodal AI

Development Tools

- AI Code Editors
- Code Completion Tools